

Knowledge Graph Neural Networks for IIoT Intrusion Detection: A Baseline Comparison Study

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Abstract

This study investigates whether graph-based learning improves intrusion detection in IIoT devices compared with classical machine learning classification methods. Using the CIC IIoT Dataset 2025 that contains benign and attack data, we first conduct an exploratory analysis of benign and attack traffic and frame the task as a binary classification, benign vs. attack, over 1-second windows. We establish an easy-to-interpret classification baseline, including Logistic Regression, Naive Bayes, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and k-Nearest Neighbors (k-NN).

We then design graph-based representations that model device-behavior relations as a bipartite knowledge graph with a structure similar to “device nodes – behavior-state nodes”, connecting IIoT device nodes to discretized behavioral-state nodes, and evaluate both heuristic graph-scoring methods and a prototype knowledge-graph neural network. Initial results showed that classical models such as k-Nearest Neighbors (accuracy = 0.922, precision = 0.972, recall = 0.829) and Logistic Regression (accuracy = 0.900, precision = 0.954, recall = 0.787) outperformed the initial graph analysis and the KG+GNN prototype (accuracy = 0.8789, precision = 0.9189, recall = 0.7629) in this setting. However, the baseline graph-analysis methods used a simpler device-to-behavior-state graph representation.

The later rich-graph GNN experiments extended this representation by adding nodes for IP, port, protocol, and time. Using this new representation, we obtained a different result: the graph-based GNN outperformed all previous models tested in this study under the current random window-level split, including k-Nearest Neighbors and baseline Logistic Regression. The strongest model was the GraphSAGE-rich architecture (accuracy = 0.997, precision = 0.997, recall = 0.997), followed by GAT-rich (accuracy = 0.962, precision = 0.956, recall = 0.950). These results suggest that adding device, behavioral-state, IP, port, protocol, and temporal nodes improved attack detection beyond the earlier simple graph design and the best classical baseline, k-NN. Other findings were that features such as packet length, address information, TCP activity, and port diversity are strong predictors of attacks; in contrast, the strongest benign predictive features show low-intensity and stable communication patterns, such as fewer packets per window, lower IP and port diversity, and lower TCP-flag activity.

Keywords: Intrusion Detection, IIoT Traffic, Classical Classification Methods, Graph-Based Machine Learning, Benign vs Attack, Neural Networks, CIC IIoT dataset 2025

1. Introduction

Industrial Internet of Things (IIoT) environments are challenging security settings: they involve a variety of devices and networking protocols, variable traffic patterns, and operational constraints that limit monitoring. These end-devices can be subject to attack by hackers; devices must be able to detect these threats to protect them, which is why intrusion detection in IIoT is often seen as a data-driven classification problem over aggregated network measurements (for example, packet intensity, packet

size, protocols, traffic, port activity, and many others). Protecting this is pivotal; that is why more robust identification and authentication methods are demanded (Jhariya et al., 2026).

The data collected in intrusion detection, often called also anomaly detection is very important and a driver of progress. Akoglu et al. (2015) describe it as the branch of the data mining which main focus is to discover odd patterns and occurrences in the dataset impacting areas such as security, finance, health care, law enforcement and many others. They also argue that anomaly detection has a vague universal definition, some describe it as an outlier within an observation that differs from the others. For Ahmad et al. (2025) The intrusions are attempts to gain access to a device or steal information without permission. For this study, we define anomaly detection as a deviation

^{*}Project repository: <https://github.com/hectorcas13/research-projects-ualbany-hector-castillo-phd/tree/main/iiot-intrusion-detection-gnn>

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from normal behavior and binary intrusion detection or binary attack detection as a detection of malicious activity in the dataset.

Having realistic benchmark datasets can have a huge impact on cybersecurity. We live in a globalized world, and this has accelerated the frequency and complexity of cyberattacks; that is the reason why intrusion detection is pivotal in the cybersecurity field (Zhong et al., 2024).

The Canadian Institute for Cybersecurity introduced DataSense: CIC IIoT Dataset 2025, motivated by the need for network data to support holistic security analysis. The dataset is collected in an IIoT laboratory with sophisticated infrastructure and 40 interconnected devices, and it is designed to support machine learning and deep learning research for cyberattack detection and classification. . It also includes 50 distinct attack types across seven categories, for evaluation across diverse threat scenarios (Canadian Institute for Cybersecurity, University of New Brunswick, 2025).

Online detection techniques are mostly statistical approaches regarding traffic features and behaviors (Zhang et al., 2010). Attack detection has shifted from rule-based systems that heavily relied on network traffic patterns already predefined to models that are more flexible, generalize better, and can learn from data for detecting ever-evolving threats. Machine learning and deep learning have proven promising and effective in many fields, such as image recognition, speech processing, and cybersecurity attack detection; they remain attractive today because they are highly interpretable, computationally efficient, and provide a strong benchmark for assessing whether more complex models will detect additional attacks or miss existing ones more than rule-based ones (Alharthi et al., 2025).

Bilot et al. (2023) also support the idea that, historically, rule-based cyberdefense tools have been widely used and integrated into antivirus software and Intrusion Detection Systems (IDS); however, they are limited by poor generalization when new attack patterns emerge.

Graph-based security modeling represents a communication as a structure. In a graph-based model, the entities such as devices, endpoints, ports, and protocols become nodes, while communications become edges. Graph Neural Networks (GNNs) have received increasing attention in intrusion detection because they can learn from both node attributes and graph topology, potentially capturing relationships that do not manifest as simple linear effects in tabular methods. Graph theory is pivotal in the cybersecurity field; using graphs, particularly adjacency and Laplacian matrices, we can analyze networks to distinguish normal traffic from attack behavior (Zhong et al., 2024; Bilot et al., 2023; Jaber et al., 2025).

Knowledge graphs introduced by Google in 2012 impose a more explicit structure on the entities and relations, which can improve explainability. It can be categorized into a general knowledge graph that focuses on breadth of knowledge and is widely used in intelligent search and rec-

ommendation scenarios, such as Freebase and Wikidata, and a domain-specific knowledge graph that prioritizes depth of knowledge and analysis requirements within a particular domain. Still, research on graphs and dataset availability is scarce (Li et al., 2024).

Bilot et al. (2023) state that Graph Neural Networks (GNNs) have gained popularity for providing useful semantic representations from graph-structured data. For intrusion detection, graphs have been used for years because they help reveal structural patterns in cyberattacks. They can achieve tremendous results for a variety of cybersecurity problems, such as threat intelligence, anomaly or attack detection, and others. Actually, most state-of-the-art approaches deal with malicious entity detection as a node classification problem and malicious flow detection as an edge classification problem, while classifying entire network graphs is less commonly used.(Bilot et al., 2023).

Gökstorp et al. (2026) suggest that deep learning methods have gained greater popularity for anomaly detection than classical machine learning methods, but emphasize that this does not imply that DL is better or vice versa; both approaches have their advantages. For instance, DL needs significantly more resources to deploy; therefore, more research in this field is encouraged.

In this study, we model the data for IIoT intrusion detection on the CIC IIoT Dataset 2025 (benign and attack data) by:

- I - Exploring a comparison between a benign and an attack behavior
- II - Modeling classification methods under testing and 10-fold cross-validation,
- III - Constructing 3 graph analysis designs as intermediate steps, with a simple device-state graph,
- IV - Implementing a GNN prototype to test improvements over non-graph models.
- V - Re-engineer the features to expand the graph to richer ones with device, behavior-state, IP, port, protocol, and time buckets information
- VI - Compare all the resulting metrics for all the models

Finally, our work will leverage the power of Codex 5.5 and Python 3.11 to generate most of the coding, plots, and analysis, and Visual Studio Code as the IDE (OpenAI, 2026; Microsoft, 2026).

2. Related Work

For this study, we sought out different approaches to detecting cybersecurity behavior that have been used in the past and compared them with the current state of the art.

2.1. Intrusion Detection Background

Scarfione et al. (2007) discuss that intrusion detection is understood as a process for monitoring the systems and network events, looking for signs of hacking incidents or

policy violations. Intrusion detection has been a research field since the late 1990s, so machine learning has been part of it since then. For two decades, it was the largest research field until the 2010s (Gökstorp et al., 2026). Nowadays, we have intrusion detection and prevention systems designed not only to identify suspicious activity but also to log and report it, and sometimes to stop such events. Foundational guidance from NIST (Scarfone et al., 2007) emphasizes that effective intrusion detection rarely relies on a single technique; instead, cybersecurity threats in corporate networks have evolved so much that right now it poses a huge challenge that requires advanced detection methods, prompting practitioners to develop, and test new techniques for example, machine learning, deep learning, graph methods, and even hybrid approaches (Hwang et al., 2026; Alharthi et al., 2025; Bilot et al., 2023). As information security attacks have become more sophisticated over time, driven by the rise of AI, new approaches have been adopted to detect them.

An early approach to filtering benign and attack behavior on a network was proposed by Zhang et al. (2010). They proposed a nonparametric CUSUM algorithm to detect real-time DDoS attacks on leaf routers by analyzing IP traffic behavior, deploying it at the entrance of the victim network, for example, a campus network. Their results showed that using a DDoS detection algorithm and the CUSUM algorithm can distinguish normal traffic from attack behavior with short detection times and high accuracy, which is very important because DDoS is the most common attack (Zhang et al., 2010; Alharthi et al., 2025; Altaf et al., 2024).

2.2. Machine Learning and Deep Learning Approaches

(Huang Jingwei et al., 2017) propose a Latent Dirichlet Allocation (LDA) solution for organizational cybersecurity, which should not be confused with Linear Discriminant Analysis (LDA). Their idea is twofold: first, to discover hidden behavioral patterns from security monitoring logs; and second, to identify sensitive information in outgoing emails to detect and help prevent data leaks. In methodological terms, this is an unsupervised machine learning approach, as his LDA is a topic modeling method that discovers latent structure in the data without relying on predefined class labels.

Another push forward in the adoption of cyber defense techniques was the advancement of cloud computing, which has posed greater challenges for enterprises and organizations in security policies, intrusion detection, and anomaly detection (Huang Jingwei et al., 2017).

A more recent study by Alharthi et al. (2025) tested various models on labeled binary and multiclass datasets, including Trees, k-NN, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), XGBoost, and others, to evaluate their performance. The authors encountered that Machine learning models such as Random Forest, Decision Tree, and XGBoost outperform the other Deep Learning

models. They also acknowledge that whole deep learning models are very good at capturing complex feature interactions, but they can suffer from data imbalance and overfitting. Another constraint is that they are computationally heavy for real-time attack detection (Alharthi et al., 2025).

2.3. Graph-Based Intrusion Detection

Bilot et al. (2023) explore various state-of-the-art techniques for intrusion detection analysis using Graph Neural Networks (GNNs). Anomaly detection has evolved from Machine Learning methods that operate on flat data, making it difficult to detect structural relational attacks, to graph-based detection techniques, which enable more efficient modeling of hosts, flows, and processes. Recent methods rely on attributed heterogeneous, spatio-temporal, and dynamic graphs because they better represent the complexity of real-time cyberattacks. Architectures like GCN, GraphSAGE, GAT, and HAN have been adapted for cybersecurity tasks. Bilot et al. (2023) conclude that GNNs are the way forward for the cybersecurity field and the attack detection exercise as they can learn structural and contextual data representations of attacks, and recognize that there are important challenges, for instance, scalability, temporal modeling (how the data changes over time), and robustness against adversarial attacks.

For botnet detection, we identified a combined approach using graph neural networks with a multi-edged graph structure that learns highly complex temporal patterns. The graph developed comprises nodes V that represent IIoT devices or IP addresses and the edges $\mathcal{E}_T$ that represent how an end-device communicates at time $T = [t_0, t_1, \dots, t_T]$. Altaf et al. (2024) also refer to this as Gated Graph Convolutional Neural Network (GGCN), which outperformed traditional current GNN approaches because it can analyze network traffic dynamically for dynamic cyberthreats in IIoT devices. This suggests that more focus should be given to enhancing this model approach (Altaf et al., 2024).

Jaber et al. (2025) introduce a technique called SPECTRA graph analysis with a focus on IIoT devices. In contrast to our dataset, where each row represents a 1-second window, they first apply a preprocessing step to create a time-series representation. Specifically, the authors aggregate the original traffic logs into 1-second intervals, so that each row in the STS representation corresponds to a 1-second time window rather than to a single raw log entry. His results with SPECTRA showed that it effectively captures patterns in flow and connectivity dynamics in start graph and bipartite topologies, outperforming other evaluated methods, such as the XGBoost classifier, whose performance varies by attack type (Jaber et al., 2025).

2.4. Hybrid Graph and Language-Based Approaches

Hwang et al. (2026) explore a combination of GNN and LLMs, called Graph-LLM. By transforming network flow

records into a bipartite graph and enriching the graph representations with contextual embeddings generated by TinyLlama. Although they achieved strong detection performance, the authors acknowledged that it is computationally expensive due to the integration of LLM embeddings (Hwang et al., 2026).

Other hybrid techniques for anomaly detection are the MABF-IoT framework. It leverages blockchains, an anomaly-detection system based on GNNs, and digital fingerprints for authentication. His approach easily outperforms other classical methods in several areas, such as accuracy, threat detection, and resource efficiency (Jhariya et al., 2026).

This aims to be a supporting study for future research on tabular methods, graph analysis, and neural networks for attack detection.

3. Dataset and Problem Definition

The data set utilized in this study was obtained from the Canadian Institute for Cybersecurity website. This is a real-time, sensor based dataset for attack analysis in IIoT devices, comprising two subsets: one with data from real attacks (attack_samples_1sec.csv) and another with data from normal behavior (benign data) (benign_samples_1sec.csv) (Canadian Institute for Cybersecurity, University of New Brunswick, 2025).

We may see the word “window” a lot in this study. This means the fixed time interval used to aggregate traffic into one row of our dataset. It is a 1-second window, so one row represents the summary of network activity observed during that single second. Every row in both datasets corresponds to a 1-second window for a device, with the device’s aggregated traffic during that interval.

3.1. Benign dataset summary

This data set contains only benign traffic. All label columns correspond to the same benign class. This dataset is useful for understanding normal behavior in IIoT devices; nevertheless, it is not enough to answer the research question, as we will need a dataset that shows attacks behavioral components to compare against. For benign dataset, we performed the following summary:

Table 1: Benign dataset summary

Metric	Value
rows	136800
columns	94
unique_IoT devices	38
unique_IoT device_MAC address	38
time_start	2025-09-09 14:09:40
time_end	2025-09-09 15:09:40
duration_minutes	60.0

The dataset has a lot of features that are almost always zero, fully constant or just garbage, those values must be

removed because it will add little predictive value or noise to our models, some examples are:

- network_fragmentation
- network_fragmented-packets
- network_tcp-flags-urg_count

A review was conducted on the dataset to see what the most common devices are. It was a lot of help as it enabled us to identify which devices are more active, learn normal behavior patterns, and define the graph entities for example: IPs, MACs, protocols, and ports.

3.2. Attack dataset summary

This file contains only attack observations. The attack labels span multiple high-level families, such as reconnaissance (recon), denial of service (DoS), distributed denial of service (DDoS), man-in-the-middle (MitM), malware, web-based attacks, and brute-force attacks, along with many subtypes of attacks such as scans, spoofing, Mirai floods, and protocol-specific flooding variants. This is very useful because it provides a substantial malicious dataset for comparing against the benign baseline. For attack dataset, we performed the following summary:

Table 2: Attack dataset summary

Metric	Value
rows	90391
columns	94
unique_IoT devices	38
unique_IoT device_macs address	38
time_start	2025-01-15 13:04:54
time_end	2025-06-12 19:11:54
duration_seconds	12809220.232

The attack dataset spans approximately 148 days, from January 15, 2025 to June 12, 2025. This does not mean that attacks occurred continuously during the entire period; it only indicates that the attack observations in the attack dataset are distributed across this timestamp range.

Table 3: Top attack patterns ranked by mean packets across the 38 unique IIoT devices.

Rank(s)	Attack Type	Port(s)	Mean Packets	Range
1, 2, 4	udp-flood	1883, 80, not port-specific	77141.72	123634.91
3, 5	rst-fin-flood	80, 1883	73574.88	92985.35
6, 8, 11, 14	syn-flood	22, 80, 1883, 443	42252.94	72460.22
7, 9	push-ack-flood	80, 1883	56628.80	59174.61
10, 12	tcp-flood	1883, 80	54775.05	56498.11
13	icmp-flood	not port-specific	50079.40	
15	ip-spoofing	not port-specific	40847.26	

Table 3 groups the original top 15 attack-port combinations into broader attack types. This makes the table easier to interpret because attacks such as udp-flood-port-1883 are separated into the attack behavior, udp-flood, and the affected port, 1883. The table summarizes the highest packet-volume attack patterns observed across the 38 unique IIoT devices in the dataset.

Many flood-style attacks have extremely large `mean_packets` and large `median_packets`, indicating sustained high-rate behavior that does not appear to be driven by outliers.

Recon-style subtypes tend to have smaller packet volumes but show higher activity on the ports relative to their packet counts. This may indicate probing behavior in which fewer packets per target are distributed across many ports.

Some devices show `median_packets` close to zero while still having a significant `mean_packets`. This relates to sparse attack windows, where many seconds contain little measurable activity, while a smaller number of seconds contain very strong bursts that increase the mean.

3.3. Data curation

The data curation process began by loading the benign and attack traffic files from the CIC IIoT Dataset 2025 downloaded from (Canadian Institute for Cybersecurity, University of New Brunswick, 2025). Each row in our dataset represents a 1-second traffic window. Also, a binary target variable was created, where benign traffic was coded as 0 and attack traffic as 1. The datasets were inspected for missing values, duplicated rows, constant numeric columns, and non-numeric fields. The resulting process from the data curation was:

- no duplicated rows or missing values
- no constant numeric features were removed
- for the tabular baseline models, only numeric traffic features were retained

Table 4: Dataset sizes before and after combining benign and attack data.

Dataset	Rows	Columns	Class
Benign data	136,800	94	Benign
Attack data	90,391	94	Attack
Combined raw dataset	227,191	94	Benign + Attack
Combined modeling dataset	227,191	96	Added dataset_class and target

Two more columns were added to the dataframe increasing the number of columns to 96. The column `dataset_class`, that contains values (benign, attack) and target that contains values for (0,1), corresponding to benign and attack binary values respectively. The created target column remains as the response variable, i.e. what we want to predict. However, we do not use it as a feature for modeling prediction.

The following columns were excluded for tabular methods:

Table 5: Features excluded from the modeling dataset.

device_name	device_mac
label_full	label1
label2	label3
label4	timestamp
timestamp_start	timestamp_end
log_data-types	network_ips_all
network_ips_dst	network_ips_src
network_macs_all	network_macs_dst
network_macs_src	network_ports_all
network_ports_dst	network_ports_src
network_protocols_all	network_protocols_dst
network_protocols_src	dataset_class

Many columns were evaluated using Python dataframes and Columns such as `label1`, `label2`, `label3`, `label4`, `label_full`, and `dataset_class` cannot be used as predictors because they don’t add any value to the models. The 24 columns were not removed because they did not add value in general, they were excluded from the tabular models because they were not numerical columns, identifiers, timestamps, lists or labels that could lead to data leaks. Some do add value for graph analysis, particularly those relating to devices, IP addresses, ports and protocols. Hence, for the tabular models 71 features were included.

3.4. Problem definition

Scarfone et al. (2007) argues that an intrusion detection system has a trade-off between false positives and false negatives. Neither can be completely eliminated, and typically, reducing one increases the other. Some companies prefer to minimize false negatives, albeit at the cost of increased false positives. Our dataset is imbalanced between benign and attack observations.

A binary supervised learning approach was used to compare different models and determine which models perform better at detecting attacks. Supervised statistical learning involves creating a statistical model to predict an output Y from inputs X . For this study, we define

$$Y \in \{0, 1\},$$

where 0 = benign and 1 = attack, with the objective of identifying features that predict both classes (James et al., 2013).

For this study, we have the following empirical question:

“Does using a Knowledge Graph and Graph Neural Network (GNN) detect attacks on IIoT better than using standard models that do not use graphs?”

To answer this question, a performance test was run to compare tabular/classification methods vs. graph analysis techniques. This question is pivotal for IIoT, where complex models are costly unless gains are clear.

Based on previous studies, such as (Hwang et al., 2026; Bilot et al., 2023; Gökstorp et al., 2026; Altaf et al., 2024), which state that the use of Graph Neural Networks is not a one-size-fits-all, but it is a strong approach in the field of cybersecurity threats detection, we formulate the following hypothesis:

“Knowledge Graph Neural Network achieves better performance for intrusion detection than classical methods in IIoT devices.”

4. Methodology

In this paper we describe the workflow used to compare classification tabular methods with Graph-based intrusion detection techniques. The study is organized into four parts: the evaluation protocol, the tabular baselines, the graph-analysis designs, and the knowledge graph with a graph neural network. Later, the rich graph representation extended this structure by adding IP, port, protocol, and temporal nodes, allowing the Neural Network models to learn from a more heterogeneous knowledge-graph-style representation. We also show the formulas from the books to better understand the theoretical and mathematical foundations of the analysis made using Python for classification to obtain our results, linking the implementation to the underlying methodology and ontology.

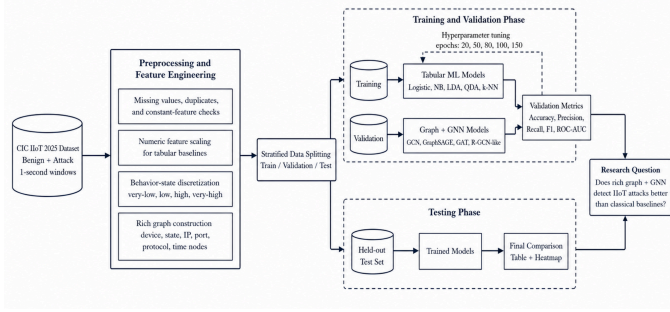


Figure 1: System model for IIoT intrusion detection using tabular baselines, Graph analysis and rich graphs GNNs - generated by Codex 5.5 (OpenAI, 2026)

4.1. Tabular classification methods

The intrusion detection study performed in this paper is a binary supervised classification problem. Each observation represents a 1-second network traffic window from the CIC IIoT 2025 dataset. The response variable is defined as:

$$y_i = \begin{cases} 0, & \text{benign traffic,} \\ 1, & \text{attack traffic.} \end{cases}$$

Each row is represented as a feature vector:

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip}),$$

where p is the number of selected traffic features after preprocessing.

The dataset was first divided using an 80/20 stratified train-test split. Then, 10-fold cross-validation was applied to the tabular baselines to preserve the benign/attack proportion across both sets. The training set was used to fit

and apply some tweaks to the models and adjust thresholds when needed, the test set was used for final comparison with all the models. Moreover, 10-fold CV (k-fold cross validation technique) was applied to the tabular baselines to evaluate if model performance is stable across different training partitions.

We used an 80/20 stratified split with `random_state = 42`.

Table 6: Training and test split distribution.

Split	Total Rows	Benign	Attack	Benign %	Attack %
Training set	181,752	109,440	72,312	60.21%	39.79%
Test set	45,439	27,360	18,079	60.21%	39.79%

Then we assessed the model using the parameters for classification for example: accuracy, precision, recall, F1-score, and ROC-AUC. In this study, recall and F1-score are especially important because missed attacks imply more money spent by companies when it comes to intrusion detection.

The evaluation metrics are defined as follows: we compare the models using the following evaluation metrics, which summarize different aspects of binary classification performance, including overall correctness, positive prediction quality, sensitivity to attacks, true negative performance, ranking ability, and probability calibration.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (4)$$

$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx \quad (6)$$

$$\text{Brier Score} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2 \quad (7)$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives. In addition, TPR denotes the true positive rate, FPR denotes the false positive rate, p_i is the predicted probability for observation i , $y_i \in \{0, 1\}$ is the true class label, and N is the total number of observations.

4.1.1. Logistic Regression

James et al. (2013) says that Logistic Regression is the standard baseline for binary classification techniques. It estimates the probability that a window belongs to the attack class. The model is defined as:

$$P(Y = 1 | \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}}$$

where β_0 we know it as the intercept and $\beta_1, \dots, \beta_p$ are the coefficients. Positive coefficients indicate features associated with higher attack probability and negative coefficients are features associated with benign behavior (James et al., 2013).

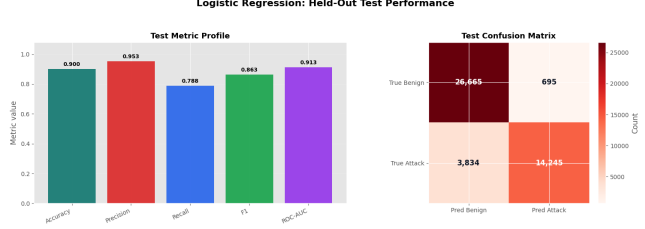


Figure 2: Held-out test performance profile for Logistic Regression

4.1.2. Naive Bayes

Naive Bayes is used to estimate the probability of each class using Bayes theorem shown below, it assumes that predictors are conditionally independent given the class label:

$$P(Y = k | \mathbf{x}) = \frac{P(Y = k)P(\mathbf{x} | Y = k)}{P(\mathbf{x})}$$

The naive assumption simplifies the likelihood as:

$$P(\mathbf{x} | Y = k) = \prod_{j=1}^p P(x_j | Y = k)$$

Although this independence assumption is often unrealistic in network traffic, Naive Bayes provides a simple useful baseline (James et al., 2013; Sahami et al., 1998). However, (Hocevar & Medkova, 2025) suggest that it may lag behind when the input distribution becomes more complex or adversarial, especially when dealing with AI-modified spam that significantly reduced its detection performance.

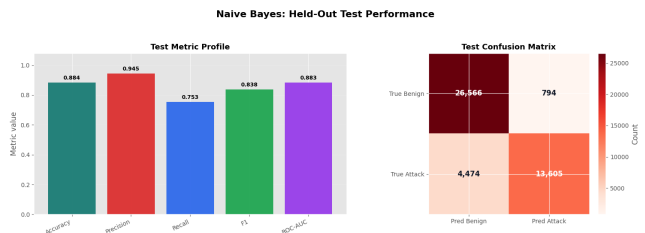


Figure 3: Held-out test performance profile for Naive Bayes

4.1.3. Linear Discriminant Analysis

Fisher (1936, as cited in Mardia, Kent, & Taylor, 2023, p. 2) (Mardia, 2023) introduced Linear Discriminant Analysis or LDA technique in 1936. LDA assumes that each class follows a multivariate normal distribution and that

the classes share the same covariance matrix. The general assumption is:

$$\mathbf{x} | Y = k \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$$

Because the covariance matrix is shared across classes, LDA produces a linear decision boundary.

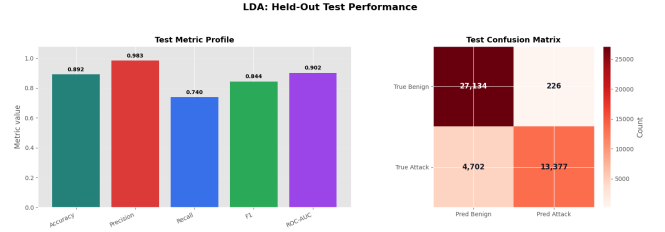


Figure 4: Held-out test performance profile for LDA

4.1.4. Quadratic Discriminant Analysis

Quadratic Discriminant Analysis or QDA is similar to LDA, but it allows each class to have its own covariance matrix, that is the difference:

$$\mathbf{x} | Y = k \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

The matrix can differ by class, that is why QDA creates a quadratic decision boundary. This makes QDA more flexible than LDA, it can also be more sensitive to noise, variance. (James et al., 2013).

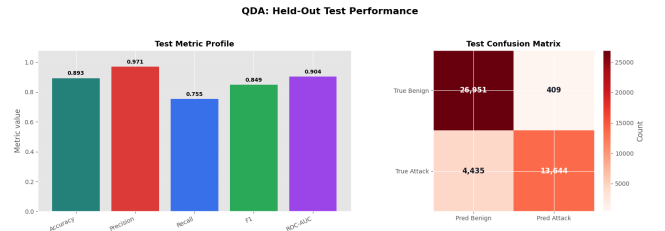


Figure 5: Held-out test performance profile for QDA

4.1.5. k-Nearest Neighbors

The k-Nearest Neighbors classifier predicts the class of a new observation based on the majority class among its k closest observations in the used training set. In this study, $k = 5$ was set as the baseline.

$\hat{y}_i$ = majority vote among the k nearest neighbors of $\mathbf{x}_i$

k-NN does not assume a specific functional form, it classifies attack and benign windows based on similarity to neighbors (James et al., 2013; Alharthi et al., 2025).

4.2. Graph Analysis Designs

This stage, in which we performed three different graph analysis, was conducted to test whether the relationships in the traffic data were meaningful to answering our research question and as a preliminary step before applying

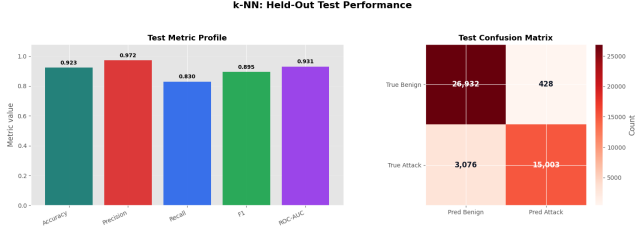


Figure 6: Held-out test performance profile for k-NN

a GNN. Instead of treating each row as a flat vector, the graphs represent devices and traffic states as connected entities. Graphs offer a richer universal representation that offers a sophisticated perspective of a system.

Unlike flat data (records in a table or in a .csv file), which strips away the relational context losing the big picture, graph capture the deep context necessary to detect anomaly behavior, as most intrusions consist of a specific interaction between host and processes; using a relational graph makes it easier to capture these patterns (Bilot et al., 2023; Jiang et al., 2019). A graph is simply represented as

$$G = (V, E)$$

where V is a set of nodes (or entities) and E is a set of edges (or relations) between those nodes.

(Bilot et al., 2023) indicate that graphs representation involves extracting informative features, they referred to this as “embeddings” from a graph. It is possible to capture different information into embeddings. Therefore, the first graph representation used in this study was a device-to-behavior-state graph. We decided to make it this way, to evaluate graph initial performance, IIoT devices were represented as device nodes, while selected numerical traffic features were discretized into behavior-state nodes such as low packet count, high port diversity, or high IP-length values. The edges connected each device to the behavior states observed in its 1-second traffic windows. This representation was used for the initial graph-analysis methods and for the first simple GCN model. This creates a bipartite graph. A bipartite graph has two types of nodes and edges only across the two groups. Here the devices are in one group and behavior are the other.

device node $\rightarrow$ behavior-state nodes

Table 7: Example flat IIoT record used to build the simple device-to-behavior-state graph.

device_name	packets_state	ports_state	ips_state
mqtt-broker	high	very-high	high

Table 7 can be represented as a graph in which the device is modeled as the central node and its related traffic attributes are modeled as connected nodes (Noble & Cook, 2003). This representation transforms a single 1-second traffic window from a flat tabular record (a table row) into

a relational structure that includes source IPs, destination IPs, packet count, destination ports, and protocols.

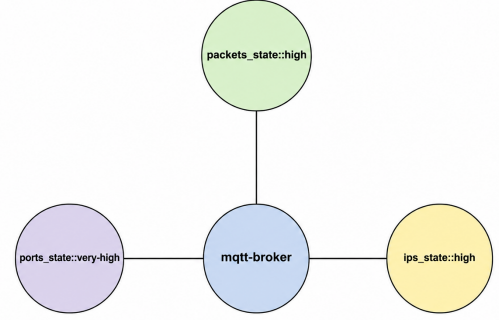


Figure: Example of the simple device-behavior-state graph representation.

Figure 7: Graphical representation of one IIoT traffic window as a device-centered graph - generated by Codex 5.5 (OpenAI, 2026).

In this representation 8, we can visualize the devices as nodes connected the behavior-states such as ports of destination (ports_dst), length of the IP (ip_length), packet count (packets_all) and others. They are categorized as very low, low, high, and very high.

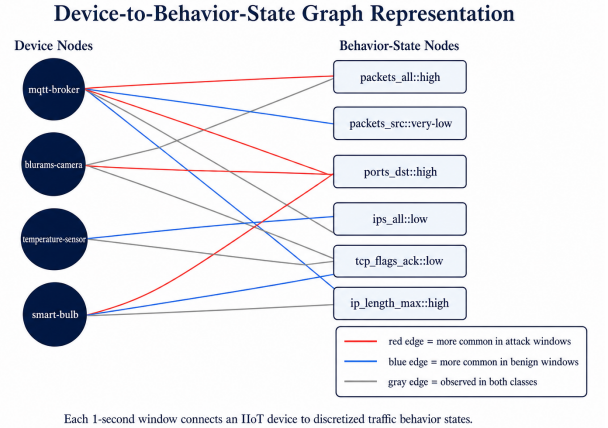


Figure 8: Example of device-state graph representation

Jiang et al. (2019) explain that a problem with many anomaly detection models is that they only consider individual entity properties, which can lead to more false positives. They argue that the connections or relationships between entities should also be taken into account, therefore using Graphs and GNN can be useful tools for anomaly detection.

4.2.1. Graph 1: Benign Novelty Graph

The first graph method used is benign traffic, which is our reference model. We believed is that attack windows

may contain device-state relationships that were not commonly observed during benign behavior. The idea is to learn normal patterns and flag as suspicious anything that rarely or never occurs in benign behavior. Noble & Cook (2003) mention that there is not a formal definition of what an anomaly detection means, but in their work they refer to it as a surprising event or an unusual occurrence. In our study, we are using a labeled dataset containing benign and attack behavior, for this initial graph we will use only the benign dataset as heuristic to capture unusual or different behavior that do not look benign. This method relates to anomaly detection techniques; it asks whether the pattern is unusual compared with benign behavior.

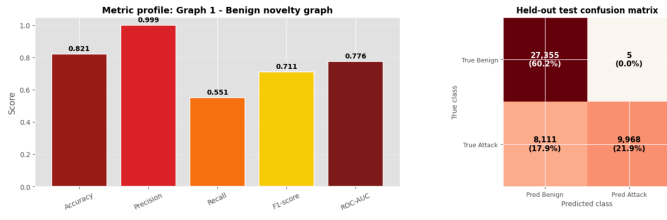


Figure 9: Held-out test performance of the Benign Novelty Graph, including evaluation metrics and confusion matrix

4.2.2. Graph 2: Dual-Class Affinity Graph

The second graph method compares each device-state relationship against both benign and attack training patterns.

We ask the following question:

do this window's device-to-state edges look more like the attack graph or the benign graph?

In this heuristic, we now use both datasets benign and attack. In Graph 1 figure 9, we only considered the benign dataset and sought for outliers in it to classify it as attack behavior. In a few words, this method is more balanced than the pure novelty detection we did previously. To do this, we use the same graph model `deviceName -> behaviorState`, for example:

```
mqtt-broker → network_packets_all_count::high
mqtt-broker → network_ports_dst_count::very-high
```

An affinity score is computed by comparing attack support and benign support with the following scoring formula:

Let a_e be the attack edge count and b_e be the benign edge count. The dual affinity score is defined as:

$$S_{\text{dual}} = \overline{\log(1 + a_e)} - \overline{\log(1 + b_e)}$$

We define `attack_edge_count` as the number of times the connection appears in attack windows, and `benign_edge_count` as the number of times it appears in benign windows, which we define as normal behavior. We

computed the Affinity-class for every 1-second window using the training set. The threshold was calculated using the F1 score, testing various possible thresholds, and the resulting threshold that gave us the best overall result was 0.044.

$$\begin{cases} \text{score} \geq 0.044 & \rightarrow \text{attack} \\ \text{score} < 0.044 & \rightarrow \text{benign} \end{cases}$$

If the resulting score is high, it indicates more attack-like behavior; A positive score indicates that the window is more structurally similar to attack traffic, while a negative score indicates stronger similarity to benign traffic.

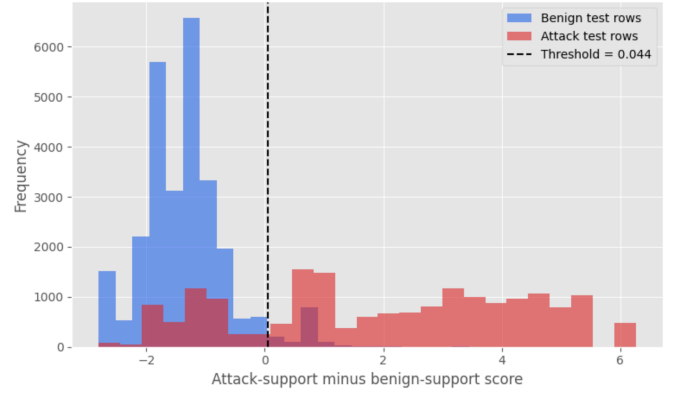


Figure 10: Dual-class affinity score distribution on the test set

After applying the dual-class affinity scoring function to the held-out test set, the results are shown in figure 11

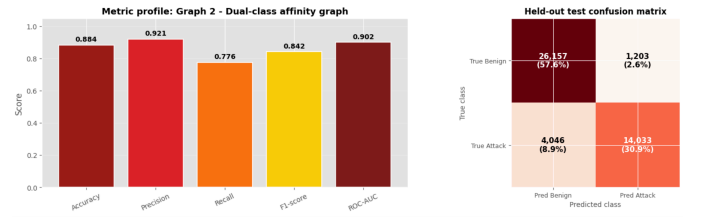


Figure 11: Held-out test performance of the dual-class affinity graph, including evaluation metrics and confusion matrix

4.2.3. Graph 3: State Co-occurrence Graph

The third graph method focuses on combinations of traffic states. Some attacks may not be identified by a single feature but by how multiple feature states co-occur. An example is a high packet count combined with high port activity; it can be more suspicious than either feature alone. As we did in Novelty Graph 9 and Dual-class Affinity Graph 11, we use the `deviceName -> behaviorState` structure. We look at which states appear together within the same 1-second window.

For example, if we see a row with the following:

network_packets_all_count::high
network_ports_dst_count::very-high
network_ips_all_count::high
network_tcp-flags-ack_count::low

We ask:

Does this combination of states occur more frequently in benign traffic or in attack traffic?

The threshold was calculated the same way as in Graph 2 as shown in figure 10, using the training set that yielded the best F1 score. We test various possible thresholds using the existing scores. We chose the threshold that best balanced precision, recall, and F1 score. The resulting threshold was -0.082.

$$\begin{cases} \text{score} \geq -0.082 & \rightarrow \text{attack} \\ \text{score} < -0.082 & \rightarrow \text{benign} \end{cases}$$

The logic was as follows:

1. We used the converted the numerical variables to very low, low, high, and very high as we did.
2. For each window, we took the resulting states.
3. We created pairs of states that appear together within that same window.
4. We counted how many times each pair appeared in the benign and attack categories.
5. We calculated a score by comparing support for the attack to support for the benign.
6. If the score exceeds a threshold (-0.082), we classify the window as an attack.
7. We use the following formula:

Let a_p be the attack pair count, b_p be the benign pair count, and P_w be the set of co-occurring state pairs in a given 1-second window. The co-occurrence score is defined as:

$$S_{co} = \frac{1}{|P_w|} \sum_{p \in P_w} [\log(1 + a_p) - \log(1 + b_p)]$$

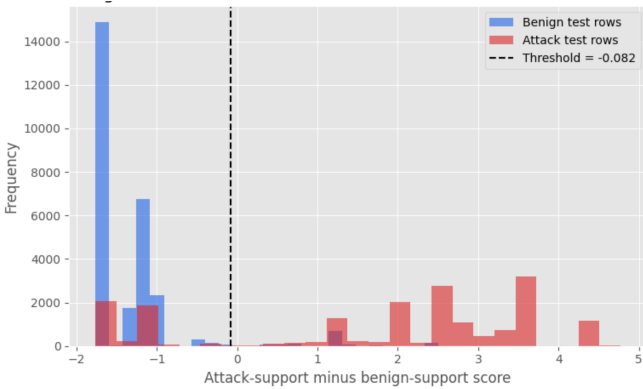


Figure 12: State co-occurrence score distribution on the test set

The main idea is that an attack behavior is not always identifiable by a single isolated variable, but rather by a combination of signals. We aim to capture a combination of states that deviate from normal behavior. That aligns with the work of Akoglu et al. (2015) regarding how to spot anomalies in the entire graph’s structure by analyzing its nodes, edges, and subgraphs. For Akoglu et al. (2015), anomaly detection is often relational and can capture real-world phenomena. It is possible to have a richer graph representation in which nodes and edges exhibit different attributes. For example, in our study, many packets, many destination ports, many IPs, many source IPs, and many protocols can be suspicious, rather than just many packets alone.

After computing the results for the State co-occurrence graph, we obtain the following results shown in figure 13

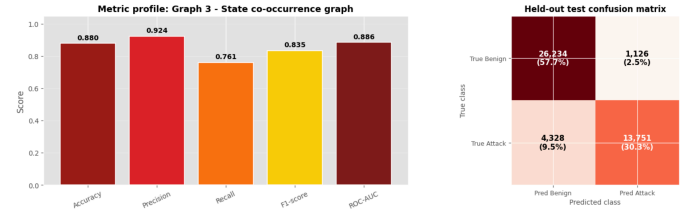


Figure 13: Held-out test performance of the state co-occurrence graph, including evaluation metrics and confusion matrix

4.3. Graph Neural Networks (GNNs)

Neural Networks, often referred to as Deep Learning (DL), are an active area of research across many fields. They rose to prominence in the 1980s. DL includes Convolutional Neural Networks, which are used for image classification where have proven to be highly effective (James et al., 2013).

Graph Neural Networks (GNNs) are also part of DL and have become a successful approach by learning useful representations from graph-structured data Bilot et al. (2023). Cybersecurity practitioners have years of experience using graph analysis for capturing structural patterns in attack data. When we are able to represent the benign behavior of the IIoT devices as graphs. Attack behavior is characterized by a sequence of suspicious and normal behavior among entities, such as the host or processes within the host system. It is possible to capture this behavior and model it with a GNN to learn patterns in these interactions and achieve strong results applicable across various cybersecurity domains, such as threat intelligence, malware detection, and more (Bilot et al., 2023).

Bilot et al. (2023, p. 49118) state that “GNN model can be applied to directed, undirected, labeled and cyclic graphs using the concept of neighbor information propagation.”

A general Graph Neural Network (GNN) updates the representation of a node by combining its own features

with information from its neighboring nodes. In this study, this idea is useful because an IIoT device or traffic-state node is not treated as isolated; instead, its representation can be influenced by the other nodes connected to it in the graph.

$$h_u = f(x_u, x_v, h_v), \quad u \in V, v \in \mathcal{N}(u)$$

where h_u is the learned representation, or embedding, of node u , x_u represents the input features of node u , x_v represents the input features of a neighboring node v , and h_v is the learned representation of that neighbor. The term $\mathcal{N}(u)$ means the set of neighboring nodes connected to u . In other words, the GNN learns the state of a node by aggregating information from its local graph neighborhood (Bilot et al., 2023).

4.3.1. Knowledge graph + Simple GNN

Now, we are going to use Graph Neural Networks, specifically Graph Convolutional Networks (GCNs), to see whether we obtain different results compared to the normal graph. For this initial KG+GNN, we will use the same simple structure we were using for the previous 3 normal graphs:

deviceName $\rightarrow$ behaviorState

GCNs are a deep learning approach used to represent Non-Euclidean data, such as social networks and graphs, and are an extension of CNNs (Jiang et al., 2019).

Following the GCN formulation presented in the Huang et al. (2025) documentation, a graph convolutional layer can be written as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

where $H^{(l)}$ is the node embedding matrix at layer l , $H^{(l+1)}$ is the updated node embedding matrix, $W^{(l)}$ is the learnable weight matrix, and σ is a nonlinear activation function. The matrix $\tilde{A} = A + I$ represents the adjacency matrix with self-loops, meaning that each node keeps part of its own information during message passing. The matrix $\tilde{D}$ is the degree matrix of $\tilde{A}$, and it is used to normalize the adjacency matrix so that nodes with many connections do not dominate the aggregation process.

This formula is equivalent to the compact GCN notation (Bilot et al., 2023):

$$H^{(l+1)} = \sigma\left(\hat{A}H^{(l)}W^{(l)}\right)$$

where the normalized adjacency matrix $\hat{A}$ is defined as:

$$\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$$

Conceptually, the device representation can be written as:

$$h_{\text{device}} = f(\text{deviceName}, \text{behaviorState})$$

We are going to train our Knowledge graph + Simple GNN with 20, 50, 80, 100, 150 epochs and analyze the best performance according to the validation F1-score. The held-out test set is used only after selecting the best validation checkpoint. Nebius Team (2025) gives us general recommendations on the amount of epochs to use in our simple GNN model according to the size of our dataset (227,191). The maximum epoch we used is 150, it does not mean we automatically decide to use 150 epochs for all the runs. Neural Networks can reach the best generalization earlier, hence, we train up to the maximum allowed epoch and keep the highest F1-score. Following this validation steps, it help to decide when the model generalizes best. To reduce randomness and improve reproducibility, experiments used fixed seeds (random_state=42, np.random.seed(42), and torch.manual_seed(42)). However, because neural-network training can still exhibit nondeterministic behavior, especially with stochastic optimization and dropout, model selection was based on validation F1 rather than a fixed epoch number.

Table 8: Hyperparameters used in the Simple GNN model.

Hyperparameter	Value
hidden_dim	32
num_layers	2
dropout	0.1
batch_size	4096
learning_rate	1e-3
optimizer	Adam
selection_metric	validation F1-score

The hidden_dim parameter defines the size of each node embedding, where each device or behavior-state node is represented by a vector of 32 learned values. The model uses two GCN message-passing layers, allowing nodes to learn from direct neighbors and indirect neighbors. A dropout value of 0.1 was used to reduce overfitting, while a batch size of 4096 defines the number of 1-second windows processed before updating the model weights. The learning rate was set to 1e-3, equivalent to 0.001, and the model was optimized using Adam. The best checkpoint was selected using the validation F1-score rather than the test set.

Table 9: Simple GNN epoch sensitivity on the held-out test set.

Max Epochs	Best Epoch	Val. F1	Acc.	Prec.	Rec.	F1
20	19	0.79	0.85	0.91	0.71	0.79
50	45	0.81	0.87	0.92	0.73	0.81
80	80	0.82	0.87	0.92	0.75	0.82
100	100	0.82	0.87	0.92	0.75	0.83
150	106	0.83	0.88	0.92	0.76	0.83

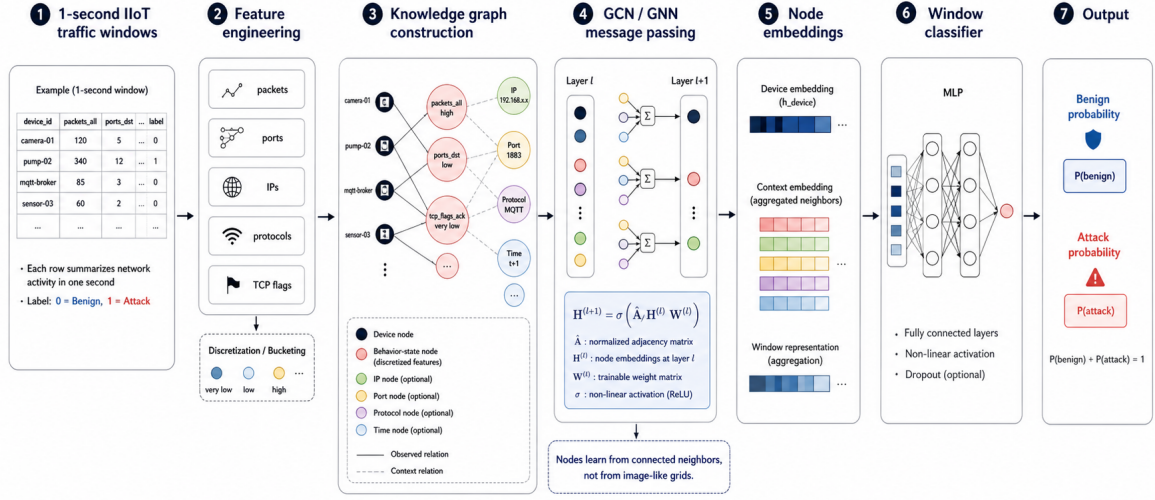


Figure 14: Architecture of the GNN-based intrusion detection workflow, from 1-second IIoT traffic windows to knowledge graph construction, message passing, node embeddings, and benign-versus-attack classification - generated by Codex 5.5 (OpenAI, 2026).

Table 10: Simple GNN confusion-matrix counts by epoch setting.

Max Epochs	Best Epoch	TN	FP	FN	TP
20	19	26021	1339	5277	12802
50	45	26184	1176	4865	13214
80	80	26115	1245	4513	13566
100	100	26107	1253	4500	13579
150	106	26143	1217	4286	13793

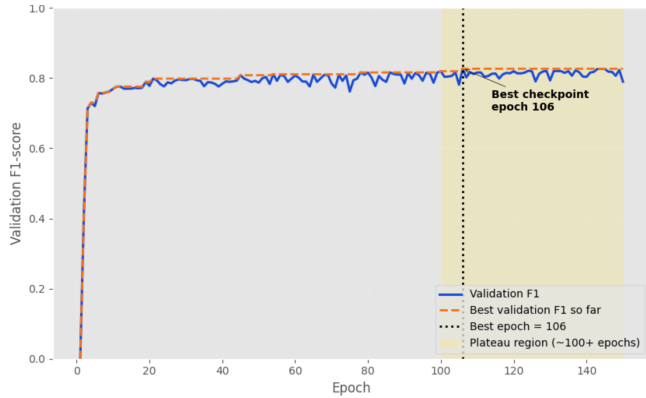


Figure 15: Simple GNN validation F1 and learning plateau

We ran several times our baseline model to get the best epoch performance and the results were close to a range between 100-150. Hence, we see that the model reaches a performance plateau after approximately 100 epochs. The exact best checkpoint may vary across runs, but it generally occurs within the plateau region between 100 and 150 epochs. In a few words, every time we ran the model with the same hyperparameters and same np.random.seeds(42)

we were obtaining a different best epoch, for instance, 100, 106, 145, therefore for this paper we chose best epoch 106, that makes sense as it is within the plateau region, see figure 15.

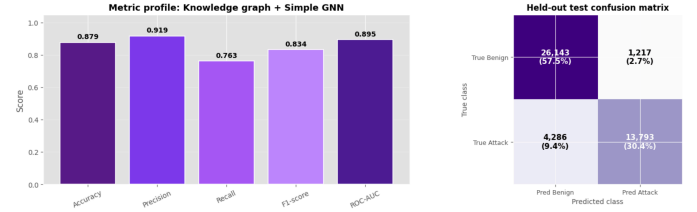


Figure 16: Graph neural network confusion matrix

Figure 14 shows the step followed to obtain the overall performance results and confusion matrix of figure 16.

4.3.2. Expansion of the Graph Structure

In the previous graph analysis, we tested a simple device-to-behavior-state. It was very useful, but comparing the results, it did not outperform the tabular methods, for instance, k-NN got a 0.92 accuracy, 0.97 in precision, and a recall of 0.83, and logistic regression got an accuracy of 0.90, precision of 0.95, and a recall of 0.79. The best graph method was the Dual Affinity-class graph in figure 11, which achieved an accuracy of 0.8845, a precision of 0.92, and a recall of 0.78. Of all the methods tested in this study, the Dual Affinity-class ranked fifth. The idea was to test if graph-based methods can outperform classical tabular methods, but our results did not show that. Therefore, based on the literature of (Altaf et al., 2024),

we decided to create a richer representation of the graph structure by expanding the initial device-to-behavior-state graph. (Altaf et al., 2024) modeled IIoT graphs containing botnet traffic and created a heterogeneous multigraph representation of network topology, with the idea of capturing the temporal dynamics of network communication across devices.

$$\begin{aligned}
G_{\text{rich}} &= (V_D, V_S, V_{IP}, V_P, V_R, V_T, E) \\
V_D &= \text{device nodes} \\
V_S &= \text{behavior-state nodes} \\
V_{IP} &= \text{IP nodes} \\
V_P &= \text{port nodes} \\
V_R &= \text{protocol nodes} \\
V_T &= \text{time nodes} \\
E &= E_{D,S} \cup E_{D,IP} \cup E_{D,P} \cup E_{D,R} \cup E_{D,T}
\end{aligned}$$

where G_{rich} is the complete graph and E means the set of edges, or the connections within the graph.

The richer graph representation in this study expands the initial device-to-behavior-state graph by adding IP, port, protocol, and temporal nodes. (Bilot et al., 2023) also supports the idea that a simple structural representation of the graph like the one we did device-to-behavior-state fails to capture patterns that are pivotal for anomaly detection. Cyberthreats now come with new attack patterns that elapse arbitrary periods of time, consequently, (Bilot et al., 2023) advocate for a richer heterogeneous representation of graphs for a better attack detection.

device $\rightarrow$ behavior-state
device $\rightarrow$ IP
device $\rightarrow$ port
device $\rightarrow$ protocol
device $\rightarrow$ time
time $\rightarrow$ next time

This allows the graph to encode not only the device behavior state, but also where the communication occurs, which ports and protocols are involved, and when the traffic window appears.

Figure 17 is a representation of the richer graph structure created to capture patterns in the dataset. To test one more time the GNNs using a richer heterogeneous graph structure in order to expand this study and answer our research question, we will compute the following three (3) GNN-rich models: GCN-rich, GraphSAGE-rich, and GAT-rich as explained in the work of (Bilot et al., 2023). For this new evaluation the same hyperparameters shown in simple GNN device-behavior-state shown in table 8, in order to make a fair comparison across the different GNNs models.

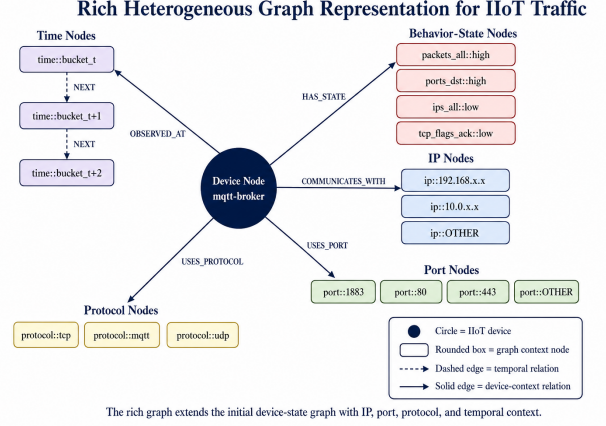


Figure 17: Rich heterogeneous graph structure connecting IIoT device nodes to behavioral states, IPs, ports, protocols, and temporal context - generated by Codex 5.5 (OpenAI, 2026).

Table 11: Node type summary for the rich graph representation.

Node Type	Count
time	7353
port	121
ip	82
protocol	41
device	38
state	33

Table 12: Edge type summary for the rich graph representation.

Edge Type	Edge Count
device_state	1744812
device_ip	302348
device_port	245276
device_protocol	196861
device_time	145401
time_next	7352

Total nodes: 7668
Total unique typed edges: 38093

Table 11 summarizes the node composition of the rich graph representation. Most nodes correspond to time nodes, followed by port, IP, protocol, device, and behavior-state nodes.

Table 12 summarizes the edge types used to connect the different node categories in the rich graph. The largest number of observed relationships corresponds to device-state edges, followed by device-IP, device-port, device-protocol, and device-time relationships. The time_next edges represent temporal continuity between consecutive time nodes.

4.3.3. GCN-rich

We use again a Graph Convolutional Neural Network (GCN), but this time with a richer graph structure. In our study, a device node learns of connected nodes such as behavior-state, IPs, ports, protocols and time nodes. It was chosen as baseline GNN because it is a classic and straightforward architecture for learning node embeddings using the graph structure.

The same formula used for simple GCN will be used for GCN-rich.

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)})$$

The GCN-rich update rule remains the same as in the simple GCN. The difference is that the adjacency matrix A is now constructed from the richer graph structure, so the model propagates information across more types of nodes and relations.

In other words, the compact formula uses $\hat{A}$ to represent the full normalized adjacency operation. In this study, A is built from the rich graph structure, including device-to-state, device-to-IP, device-to-port, device-to-protocol, device-to-time, and time bucket nodes.

After computing the results of the GCN-rich, we obtain the following result:

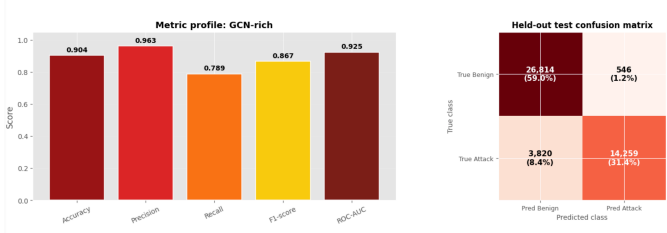


Figure 18: Rich-graph GNN test profile: GCN-rich

4.3.4. GraphSAGE-rich

GraphSAGE stands for Graph Sample and Aggregate. It is another method widely used in anomaly detection for IIoT devices. To use it for anomaly detection, when building the graph, the nodes are identified by an IP address and a port, the edges represent the communication between two nodes. It can be scaled up when new nodes added such as new devices, IPs or ports. GraphSAGE works mainly with node features and it used to learn embeddings of nodes Bilot et al. (2023).

GraphSAGE was used as a second rich-graph GNN model because it is designed to learn node embeddings by sampling and aggregating information from a fixed number of neighboring nodes. Unlike a standard GCN, which uses the full adjacency matrix, GraphSAGE samples a subset of neighbors, making it more scalable for larger graphs. In this study, this is useful because the rich graph contains multiple node types, including devices, behavior states, IPs, ports, protocols, and time nodes.

Following the GraphSAGE formulation described by (Bilot et al., 2023), the sampled neighborhood of node u is defined as:

$$\mathcal{N}_s(u) = \text{SAMPLE}(\mathcal{N}(u), s)$$

where $\mathcal{N}(u)$ is the full neighborhood of node u , s is the number of sampled neighbors, and $\mathcal{N}_s(u)$ is the sampled neighborhood.

The neighborhood aggregation step is defined as:

$$h_{\mathcal{N}_s(u)}^{(l)} = \text{AGGREGATE}^{(l)}(\{h_v^{(l-1)}, \forall v \in \mathcal{N}_s(u)\})$$

where $h_{\mathcal{N}_s(u)}^{(l)}$ is the aggregated representation of the sampled neighbors of node u , and $h_v^{(l-1)}$ is the embedding of neighbor node v from the previous layer.

The node update step is then defined as:

$$h_u^{(l)} = \sigma(W^{(l)} \cdot [h_u^{(l-1)}, h_{\mathcal{N}_s(u)}^{(l)}])$$

where $h_u^{(l)}$ is the updated embedding of node u , $W^{(l)}$ is a learnable weight matrix, σ is a nonlinear activation function, and $[\cdot, \cdot]$ denotes concatenation. After running the GraphSAGE GNN model, we obtain the following results:

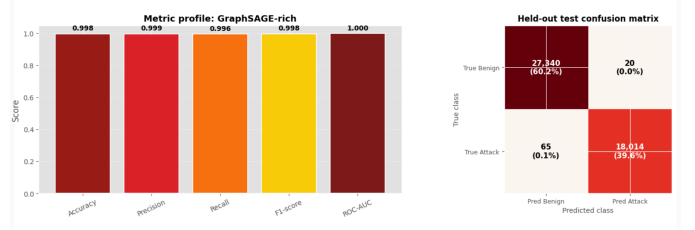


Figure 19: Rich-graph GNN test profile: GraphSAGE-rich

4.3.5. GAT-rich

GAT stands for Graph Attention Network. This GNN model can be used when the graph structure is evolving as it is very useful to generalize across diverse datasets and complex classification scenarios (Altaf et al., 2024; Ahmad et al., 2025). However, Bilot et al. (2023) say that can be unusable when working with large graphs billions of nodes or edges due to storage constraints. It stores in memory the whole adjacency matrix with corresponding features.

GAT uses an attention mechanism to assign a different importance scores to each neighbor of a node. Instead of aggregating all neighboring nodes with the same weight, GATs learn which neighbors are more relevant for the task. This allows the model to focus on more informative connections and produce a more refined aggregation of neighborhood information (Bilot et al., 2023).

$$\alpha_{uv} = \frac{\exp(\sigma(a^T [Wh_u, Wh_v]))}{\sum_{k \in \mathcal{N}(u)} \exp(\sigma(a^T [Wh_u, Wh_k]))}$$

the nodes update like:

$$h'_u = \sigma \left(\sum_{v \in \mathcal{N}(u)} \alpha_{uv} W h_v \right)$$

In this formula, α_{uv} is the attention coefficient between node u and its neighbor v . The matrix W is a shared learnable weight matrix, a is a learnable attention vector, h_u and h_v are the node embeddings, and $\mathcal{N}(u)$ is the neighborhood of node u . The final embedding h'_u is computed as a weighted aggregation of the neighboring node embeddings, where more relevant neighbors receive higher attention weights (Bilot et al., 2023).

We decided to use this model because in IIoT traffic, not all the connections are equally important. For instance, a specific port, IP or protocol may be more relevant than another when it comes to detecting attacks.

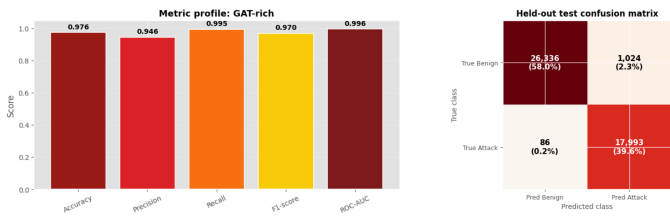


Figure 20: Rich-graph GNN test profile: GAT-rich

5. Results

For the results section, we first compare the tabular models to identify which model performed best overall. Second, we analyze the graph-based models and the Simple GNN using the initial device-behavior-state structure. Third, we evaluate the GNN models using the richer graph structure. Finally, we compare all models together and discuss their overall performance.

5.1. Comparison of classical methods

Our results show that the strongest performances in classical tabular methods are k-Nearest Neighbors (k-NN) with a precision of 0.98, recall of 0.84 and a F1-score of 0.90 and Logistic Regression with a precision of 0.96, recall of 0.78 and a F1-score of 0.86. k-NN as shown in previous studies is effective for generalization, showing high detection rates and low false positives (Alharthi et al., 2025). Other tabular methods such as LDA show strong classification overall based on the parameters. The worst was Naive Bayes for very little. These results show that classical tabular methods can still be useful for attack detection.

Table 13: Train and test performance of tabular baseline models.

Model	Split	Precision	Recall	F1	ROC-AUC
Logistic Regression	Train	0.96	0.78	0.86	0.91
Logistic Regression	Test	0.95	0.79	0.86	0.91
Naive Bayes	Train	0.95	0.75	0.83	0.88
Naive Bayes	Test	0.94	0.75	0.84	0.88
LDA	Train	0.99	0.73	0.84	0.90
LDA	Test	0.98	0.74	0.84	0.90
QDA	Train	0.97	0.75	0.84	0.90
QDA	Test	0.97	0.75	0.85	0.90
k-NN	Train	0.98	0.84	0.90	0.95
k-NN	Test	0.97	0.83	0.90	0.93

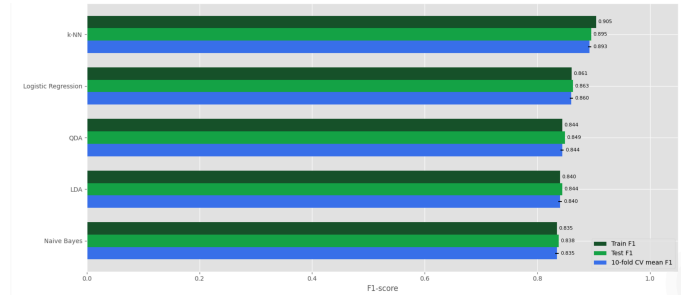


Figure 21: Train vs test vs 10-fold CV F1-score by model

5.2. Graphs analysis

We cannot discard the graph entirely; the graph analysis methods and the Knowledge Graph + GNN prototype captured meaningful signals, but overall, they did not surpass the best tabular model on the held-out test set. Among the graph-based approaches, Graph 2 (Dual-Class Affinity Graph) produced the strongest result, as shown in Table 14, indicating that comparing attack support against benign support is more effective than relying solely on simple state co-occurrence.

Table 14: Performance comparison of graph-analysis methods and Simple GNN.

Model	Metric	Value
Graph 2 – Dual-class affinity graph	Accuracy	0.88
	Precision	0.92
	Recall	0.78
	F1	0.84
	ROC-AUC	0.90
Graph 3 – State co-occurrence graph	Accuracy	0.88
	Precision	0.92
	Recall	0.76
	F1	0.83
	ROC-AUC	0.89
Knowledge graph + Simple GNN	Accuracy	0.88
	Precision	0.92
	Recall	0.76
	F1	0.83
Graph 1 – Benign novelty graph	ROC-AUC	0.90
	Accuracy	0.82
	Precision	1.00
	Recall	0.55
	F1	0.71
	ROC-AUC	0.78

We can see that Benign novelty graph reached a strong result of 1.00 precision. This means that when the model predicts attack, it is almost always right, however, the low recall of 0.55 also means that tends to miss most attacks. The best method taking into account the overall result is Dual-class affinity graph with an accuracy of 0.88, precision of 0.92 and a F1-score of 0.84. Simple structure device-behavior-state GCN did not make a good improvement overall.

5.3. Exploring richer GNNs

After the graph expansion to a richer graph structure, our model obtained more attack detecting power as shown in table 15. The best GNN was GraphSAGE-rich that reached an almost perfect overall result (accuracy = 0.9979, precision = 0.9976, recall = 0.9970, F1 = 0.9973). The second model GNN model with the best performance was GAT-rich (accuracy = 0.9629, precision = 0.9562, recall = 0.9502, F1 = 0.9532). We cannot discard GCN-rich (accuracy = 0.9054, precision = 0.8787, recall = 0.8842, F1 = 0.8815), that also showed high efficiency for attack detection.

Table 15: Rich-graph GNN model comparison.

Model	Metric	Value
GraphSAGE-rich	Best epoch	100
	Accuracy	0.9979
	Precision	0.9976
	Recall	0.9970
	F1-score	0.9973
	ROC-AUC	0.9998
GAT-rich	Best epoch	100
	Accuracy	0.9629
	Precision	0.9562
	Recall	0.9502
	F1-score	0.9532
	ROC-AUC	0.9879
GCN-rich	Best epoch	150
	Accuracy	0.9054
	Precision	0.8787
	Recall	0.8842
	F1-score	0.8815
	ROC-AUC	0.9374

The confusion matrix table below, we can observe that GraphSAGE-rich and GAT-rich are the best GNN models. For instance, GraphSAGE-rich is strong for detecting attacks (18,025) in IIoT devices and misses a few attacks (54).

Table 16: Rich-graph GNN confusion-matrix counts.

Model	TN	FP	FN	TP
GraphSAGE-rich	27,317	43	54	18,025
GAT-rich	26,574	786	901	17,178
GCN-rich	25,154	2,206	2,093	15,986

After several runs, the best epochs were something in between 80 and 140 as shown on the plateau region in figure 22.

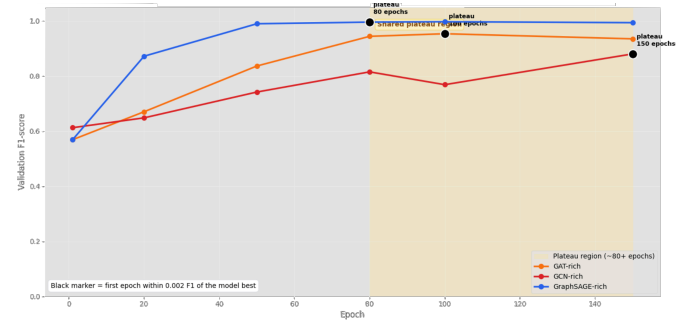


Figure 22: Validation F1 learning plateau by rich-graph GNN model

5.4. Compare against previous methods

Our research question was if GNN can have a good performance over tabular classical methods? Under the current random window-level split, rich-graph GNNs outperformed the tabular baselines.

Table 17: Overall model performance comparison.

Model	Acc.	Prec.	Rec.	F1	ROC-AUC	TN	FP	FN	TP	MAR
GraphSAGE-rich	0.9979	0.9976	0.9970	0.9973	0.9998	27317	43	54	18025	0.0030
GAT-rich	0.9629	0.9562	0.9502	0.9532	0.9879	26574	786	901	17178	0.0498
k-NN	0.9229	0.9723	0.8299	0.8954	0.9310	26933	427	3075	15004	0.1701
GCN-rich	0.9054	0.8787	0.8842	0.8815	0.9374	25154	2206	2093	15986	0.1158
Logistic Regression	0.9005	0.9541	0.7878	0.8630	0.9129	26675	685	3836	14243	0.2122
QDA	0.8934	0.9709	0.7547	0.8492	0.9037	26951	409	4435	13644	0.2453
LDA	0.8915	0.9834	0.7399	0.8445	0.9018	27134	226	4702	13377	0.2601
Graph 2	0.8845	0.9210	0.7762	0.8424	0.9018	26156	1204	4046	14033	0.2238
Naive Bayes	0.8841	0.9449	0.7525	0.8378	0.8826	26567	793	4475	13604	0.2475
Graph 3	0.8800	0.9243	0.7606	0.8345	0.8857	26234	1126	4328	13751	0.2394
Simple GNN	0.8789	0.9189	0.7629	0.8337	0.8952	26143	1217	4286	13793	0.2371
Graph 1	0.8214	0.9995	0.5514	0.7107	0.7756	27355	5	8110	9969	0.4486

Note. MAR = missed attack rate.



Figure 23: Final heatmap comparing all models across classification metrics.

Table 17 compares all models using classification metrics, confusion-matrix counts, and the missed attack rate. Figure 23 provides a visual comparison of all models across accuracy, precision, recall, F1-score, and ROC-AUC. The heatmap confirms that GraphSAGE-rich achieves the strongest overall performance, with near-perfect values across all metrics. GAT-rich is the second strongest model and also outperforms the classical baselines in most metrics. Among the classical models, k-NN remains the best-performing non-graph approach. The graph heuristics and the Simple KG+GNN capture useful signal, but they remain below the rich graph GNN models. Graph 1 shows the highest precision, but its low recall indicates that it misses many attack windows. Overall, the heatmap supports the conclusion that the richer graph representation improves intrusion detection performance when combined with stronger GNN architectures.

6. Discussion and future work

For future work in this field, one could explore combining complementary detection behaviors from multiple approaches rather than selecting a single best model. A high-recall configuration can be used to minimize missed attacks, a high-precision configuration to reduce false positives, and low-confidence cases can be routed to an intermediate reviewer category for human inspection. Adjusting thresholds can trade off missed attacks against false positives.

Another recommendation is to evaluate graph models separately by attack family and by device type, rather than only using a single combined binary classifier. This would make it possible to determine whether graph-based learning performs consistently across devices and attack categories, or whether its performance is driven by specific device-attack patterns.

Although this study compares several classical baselines with graph-based approaches, other machine learning methods, such as Random Forest and XGBoost, were outside the scope of this analysis. These methods will be considered in future research, either as an expansion of the present work or in experiments using another dataset. One example could be creating a 3rd class adjusting the threshold on the graphs, in figure 12 the resulting threshold was -0.082 but could in an expansion of this research create a review zone to review the uncertain for better generalization:

$$\begin{cases} \text{score} < -0.082 & \rightarrow \text{benign} \\ -0.082 \leq \text{score} \leq 0.082 & \rightarrow \text{uncertain} \\ \text{score} > 0.082 & \rightarrow \text{attack} \end{cases}$$

Further analysis is recommended on a similar dataset with nearly equal proportions of benign and attack data to train the models. A more balance dataset could improve performance of classical tabular methods.

We could also test graph models separately by device and attack type to verify whether performance generalizes across IIoT scenarios.

- by attack type
- by device
- by device-attack combination

For instance, an attack performed on a mqtt-broker device can contain different patterns of ports and protocols than an attack on a camera or a sensor.

This study focuses on predictive performance and graph representation design. Concepts such as entropy-based traffic characterization, notebook latency, and neural-network execution time were outside the scope of the current analysis. These aspects may be considered in future work, especially for evaluating computational efficiency and real-time deployment feasibility.

Another method that could also been taken in consideration is Ridge and Lasso. These methods apply penalties on the feature that do not value to the model, for a more efficient feature selection.(James et al., 2013). Another technique is wrapper method that evaluates subsets of features based on model performance for datasets with large volumes of data (Ahmed et al., 2025).

7. Limitations

7.1. Proportion of the dataset

The benign and attack classes are not perfectly balanced in our combined dataset (Benign: 136,800 observations; Attack has 90,391 observations; approximately 60/40). While it can be considered this a moderate imbalance overall, it can be worse at the device level and may bias models toward the majority patterns, for example, detecting more benign behavior than attack, affecting precision, sensitivity, ROC-AUC, and threshold selection. We therefore used stratified splits and k-fold cross-validation to mitigate this issue. Akoglu et al. (2015) says that unbalanced data is a big challenged since anomalies in the data are very rare and represent a smaller fraction of the whole observation.

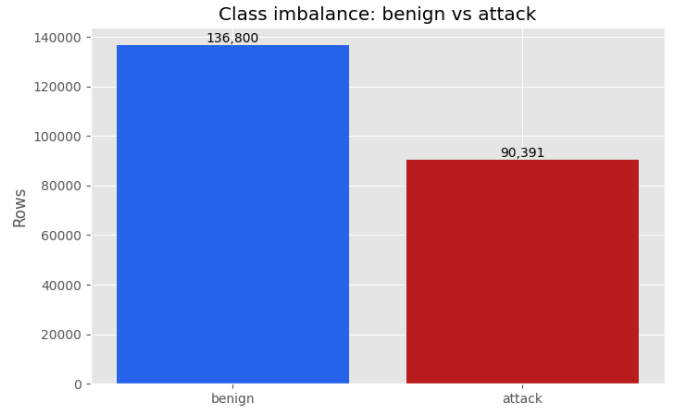


Figure 24: Class imbalance.

7.2. Computational constraints

Computing power available to make my work more efficient was critical limitation.

8. Conclusion

In this study, we evaluated whether graph-based intrusion detection analysis using a knowledge graph and a graph neural network (GNN) can outperform tabular models for IIoT attack detection on CIC IIoT 2025 1-second windows.

Our analysis showed that benign traffic is typically sparse and can become bursty, depending on the traffic observed on the device. Attack traffic exhibits much higher

packet activity, is more bursty, and shows distinct structural patterns in the plots presented.

Graph Neural Networks exhibited the strongest performance for attack detection after enriching the graph structure, as they captured more attacks than the highest-performing tabular method, k-NN. Classical tabular methods remain relevant because they are easier to interpret and less time-consuming to compute. GNNs achieved the best predictive performance in this study, but they were computationally heavier than classical methods. With large datasets, GNNs can be effective, but they may require substantial memory to run.

Declaration

AI tools were used to assist with code generation, plotting, drafting, and formatting; all outputs, analyses, and interpretations were reviewed by the author.

Grammarly was also used for grammar correction following these steps: I wrote the text in Microsoft word, Grammarly correct and recommends grammar adjustments, then we pasted the text in Codex to generate the Latex code structure in Overleaf and generate the formatting for this paper.

Declaration of Competing Interest

The author declares that he does not have known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We used publicly available datasets from (Canadian Institute for Cybersecurity, University of New Brunswick, 2025)

The notebooks are uploaded to: <https://github.com/hectorcas13/research-projects-ualbany-hector-castillo-phd/tree/main/iiot-intrusion-detection-gnn>

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